$\boxed{\text{Mar 12 2026}}$

LTV:

$\dot{x} = A(t)\, x \qquad x(t_0) = x_0$

$x(t) = \underbrace{\Phi(t, t_0)}_{\text{state-transition matrix}}\, x(t_0)$

LTI: recall

$x(t) = \underbrace{e^{A(t-t_0)}}\, x(t_0)$

$\Phi(t, t_0) \equiv \Phi(t - t_0)$

LTV: $\dot{x} = A(t)\, x$

Suppose: $\dot{P}(t) = A(t)\, P(t)$

$P(t)$ is non-singular $\quad \forall t \geq t_0$

$\Phi(t, t_0) = P(t)\, P^{-1}(t_0)$

$\Phi^{-1}(t, t_0) = \left[P(t)\, P^{-1}(t_0)\right]^{-1}$

$= P(t_0)\, P^{-1}(t)$

$= \Phi(t_0, t)$

LTI: $\dot{x} = Ax$

$\dot{P} = AP$

$P = e^{At} \qquad\qquad P = 2e^{At}$

$\Phi(t, t_0) = e^{At}\, e^{-At_0}$

$= e^{A(t-t_0)}$

$\Phi(t_0, t_0) = P(t_0)\, P^{-1}(t_0) = I$

$\frac{\partial}{\partial t} \Phi(t, t_0) = \frac{\partial}{\partial t}\left[P(t)\, P^{-1}(t_0)\right]$

$= \frac{\partial P(t)}{\partial t}\, P^{-1}(t_0)$

$= A(t)\, P(t)\, P^{-1}(t_0)$

$\frac{\partial}{\partial t} \Phi(t, t_0) = A(t)\, \Phi(t, t_0)$

$$\frac{\partial}{\partial t_0} \Phi(t,t_0) = \frac{\partial}{\partial t_0} \left[P(t) P^{-1}(t_0) \right]$$

$$= P(t) \frac{\partial}{\partial t_0} P^{-1}(t_0)$$

$$P^{-1}(t_0) = Q(t_0)$$

$$P(t_0) Q(t_0) = I$$

$$\frac{d}{dt_0} \left(P(t_0) Q(t_0) \right) = 0$$

$$P(t_0) \frac{d}{dt_0} Q(t_0) + \frac{d}{dt_0} P(t_0) Q(t_0) = 0$$

$$\frac{d}{dt_0} Q(t_0) = - P^{-1}(t_0) A(t_0) \underbrace{P(t_0) Q(t_0)}_{I}$$

$$= - P^{-1}(t_0) A(t_0)$$

$$\frac{\partial}{\partial t_0} \Phi(t,t_0) = - P(t) P^{-1}(t_0) A(t_0)$$

$$\frac{\partial}{\partial t_0} \Phi(t,t_0) = - \Phi(t,t_0) A(t_0)$$

$\dot{x} = A(t)\, x$

$V(t,x) = x^T P(t)\, x$

Recall sufficient conditions:

- $w_1(x) \le V(t,x) \le w_2(x)$

  Suppose $k_1 I \le P(t) \le k_2 I, \quad k_1, k_2 > 0$

  $\Rightarrow w_1(x) = k_1 x^T x$

  $w_2(x) = k_2 x^T x$

- $\dot{V} = \frac{d}{dt}\left[x^T P(t)\, x \right]$

  $= \dot{x}^T P(t)\, x + x^T \dot{P}(t)\, x + x^T P(t)\, \dot{x}$

  $= x^T A^T(t) P(t)\, x + x^T \dot{P}(t)\, x + x^T P(t) A(t)\, x$

  $= x^T \left[A^T P + \dot{P} + PA \right] x$

  $\Uparrow$

  $\dot{P}(t) + A^T(t) P(t) + P(t) A(t) = -Q(t)$

  $w_3(x) = k_3 x^T x \Leftarrow k_3 > 0$ | $Q(t) \ge k_3 I$, $-Q(t) \le -k_3 I$

Converse

- Suppose $x = 0$ is uniformly GAS
- $Q(t)$ is continuous and symmetric
- $\exists\, k_3, k_4 > 0 \quad$ s.t. $\quad 0 \le k_3 I \le Q(t) \le k_4 I$

$\Rightarrow \exists$ symmetric $P(t)$ s.t.

- $\dot{P}(t) + A^T(t) P(t) + P(t) A(t) = -Q(t)$
- $0 < k_1 I \le P(t) \le k_2 I$

$\Downarrow$

Recall (LTI) : $P = \int_0^\infty e^{A^T \tau} Q\, e^{A \tau}\, d\tau$

LTV : $P(t) = \int_t^\infty \Phi^T(\tau, t)\, Q(\tau)\, \Phi(\tau, t)\, d\tau$

Backstepping

Certain class of

$$\dot{x} = f(x) + g(x)u \qquad x \in \mathbb{R}^n$$

Scalar $\quad \dot{x} = \underbrace{f(x)}_{\text{Scalar}} + \underbrace{g(x)}_{\text{Scalar}} \underbrace{u}_{\text{Scalar}} \qquad x \in \mathbb{R}$

$$f(x) + g(x)u = -kx \qquad k > 0$$

$$u(x) = -\frac{kx + f(x)}{g(x)} \qquad g(x) \neq 0$$

Ex. $\quad \dot{x}_1 = \underbrace{x_1^2}_{f(x)} + \underbrace{u}_{g(x)=1}$

$$\therefore u = -(kx_1 + x_1^2)$$

$$\dot{x}_1 = x_1^2 + x_2$$

$$\dot{x}_2 = u$$

$$u \equiv u(x_1, x_2) \quad \Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \qquad \text{GAS?}$$

Backstepping

Treat x_2 as "virtual" input for the x_1 sub-system $\qquad V(x_1) = \frac{1}{2}x_1^2$

$$\alpha(x_1) = -k_1 x_1 - x_1^2 \qquad k_1 > 0$$

$$z_2 = x_2 - \alpha(x_1) = x_2 + k_1 x_1 + x_1^2$$

$$\dot{z}_2 = \dot{x}_2 - \dot{\alpha}$$

$$= u - \dot{\alpha}$$

Choose $u = \dot{\alpha} - \frac{\partial V_1}{\partial x_1} - x_1 z_2$

(backstepping) $= -k_1 \dot{x}_1 - 2x_1\dot{x}_1 - x_1 - k_2 z_2$

$$= -k_1(x_1^2 + x_2) - 2x_1(x_1^2 + x_2) - x_1 - k_2 z_2$$

$$\tilde{V} = V_1(x_1) + \frac{1}{2}z_2^2$$

$$= \frac{1}{2}x_1^2 + \frac{1}{2}z_2^2$$

$$\dot{\tilde{V}} = \frac{\partial V_1}{\partial x_1}\dot{x}_1 + z_2\dot{z}_2$$

$$\dot{x}_1 = x_1^2 + x_2$$

$$= x_1^2 + z_2 + \alpha$$

$$= -k_1x_1 + z_2$$

$$\dot{\tilde{V}} = x_1(-k_1x_1 + \dot{z}_2) + z_2(-x_1 - k_2z_2)$$

$$= -k_1x_1^2 - k_2z_2^2$$

$$\Rightarrow \quad x_1 \to 0$$

$$z_2 \to 0$$

$$x_2 - \alpha(x_1) \to 0$$

$$x_2 \to \alpha(x_1)$$

$$\Rightarrow \quad x_2 \to 0 \qquad (\because \alpha(x_1) \to 0)$$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \text{ is GAS}$$

The following is not always true:

$$\dot{x} = f(x, \theta(t))$$

$$\theta(t) \to \theta^* \quad ; \quad f(x, \theta^*) = -x$$

$$\Rightarrow \quad x \to 0 \quad \times$$

$$\text{if} \quad \theta(t) \equiv \theta^* \quad ; f(x, \theta(t)) \equiv f(x, \theta^*)$$

$$\equiv -x$$

$$\dot{x} \equiv -x$$

$$\Rightarrow \dot{x} \to 0$$

Will not extend: $\dot{x}_1 = x_1^2 + x_2 + x_3$

$$\dot{x}_2 = x_2 + x_3$$

$$\dot{x}_3 = x_3 + u$$